

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-104***

“Deepwater-Penns Grove 69 kV

September 2017

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The Interconnection Customer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Chemours Company FC, LLC, the Interconnection Customer (IC), has proposed a 19.8 MW (7.5 MW Capacity) solar generating facility to be located in Pennsville Township, Salem County, New Jersey. PJM studied the AC1-104 project as an injection into the Atlantic City Electric Company (ACE) transmission system at a tap of the Churchtown-Pennsgrove 69 kV circuit and evaluated it for compliance with reliability criteria for summer peak conditions in 2020. The planned in-service date, as requested by the IC, is December 15, 2018. This date may not be not attainable due to additional required PJM studies and the Transmission Owner construction schedule.

Point of Interconnection

The Interconnection Customer requested a transmission level Point of Interconnection (POI). The AC1-104 project will connect with the ACE transmission system at a new 69 kV three breaker ring bus substation to be constructed adjacent to the Churchtown-Pennsgrove 69 kV circuit (see Attachment 1).

Transmission Owner Scope of Work

Substation Interconnection Estimate

Scope: Construct a three-position 69 kV ring bus along Line 0710 between Churchtown Substation and Pennsgrove Substation. ACE will require a land area of approximately 1 –

1.5 acres to construct a 69 kV three position ring bus. The substation will include two dedicated line terminals for Line 0710, one position for the interconnection to AC1-104, and all the required equipment (circuit breakers, control house, relays) to operate the substation.

Estimate: \$7,396,408

Construction Time: 24 Months

Required Relaying and Communications

New protection relays are required for the new terminal. An SEL-487B will be required for primary protection and an SEL-387 will be required for back-up protection. One 24" relay panel for each line terminal will be required for front line and back-up protection.

An SEL-451 relay on a 10" breaker control panel will be required for the control and operation of each new 69 kV circuit breaker.

The project will require re-wiring and adjustment of existing relay schemes at affected substations to accommodate the new 69 kV arrangement.

The cost of the required relay and communications is included in the Substation Interconnection Estimate.

Metering

Three phase 69 kV revenue metering points will need to be established. ACE will purchase and install all metering instrument transformers as well as construct a metering structure. The secondary wiring connections at the instrument transformers will be completed by ACE's metering technicians. The metering control cable and meter cabinets will be supplied and installed by ACE. ACE will install conduit for the control cable between the instrument transformers and the metering enclosure. The location of the metering enclosure will be determined in the construction phase. ACE will provide both the Primary and the Backup meters. ACE's meter technicians will program and install the Primary & Backup solid state multi-function meters for each new metering position. Each meter will be equipped with load profile, telemetry, and DNP outputs. The IC will be provided with one meter DNP output for each meter. ACE will own the metering equipment for the interconnection point, unless the IC asserts its right to install, own, and operate the metering system.

The IC will be required to provide a POTS (Plain Old Telephone Service) line or other equivalent technology approved by ACE within three feet of each Interconnection Customer's metering position to provide ACE with remote meter interrogation.

It is the IC's responsibility to send the data that PJM and ACE requires directly to PJM. The IC will grant permission for PJM to send ACE the following telemetry that the IC sends to PJM: real time MW, MVAR, volts, amperes, generator status, and interval MWH and MVARH.

The estimate for ACE to design, purchase, and install metering as specified in the aforementioned scope for metering is included in the Substation Interconnection Estimate.

Interconnection Customer Scope of Direct Connection Work

The Interconnection Customer is responsible for all design and construction related to activities on their side of the Point of Interconnection. Site preparation, including grading and an access road, as necessary, is assumed to be by the IC. Route selection, line design, and right-of-way acquisition of the direct connect facilities is not included in this report, and is the responsibility of the IC. The IC is also required to provide revenue metering and real-time telemetering data to PJM in conformance with the requirements contained in PJM Manuals M-01 and M-14 and the PJM Tariff.

ACE Interconnection Customer Scope of Direct Connection Work Requirements:

- ACE requires that an IC circuit breaker is located within 500 feet of substation to facilitate the relay protection scheme between ACE and the IC at the Point of Interconnection (POI).

Inverter Requirements

This section contains setting requirements for the operation of the IC's proposed equipment.

The inverter at the DG location shall have the following capabilities:

- Voltage flicker reduction through dynamic VAR response
- Ramp rate control
- SCADA communications
- Curtailment or other mitigation ability if high voltage were to occur
- Low voltage and system disturbance ride through
- Ability to receive and respond to a transfer trip or SCADA signal

The inverter shall operate in accordance with the IEEE 1547 series of standards that have been approved. While inverters should be capable of voltage stabilization thru dynamic VAR response and capable of low voltage and system disturbance ride through, neither of these capabilities shall be implemented until such time that the IEEE 1547 series of standards are revised and approved to include standards for these capabilities. At such time as these revised standards become available, the PV owner/operator shall cooperate with the Company to implement these capabilities with settings acceptable to the Company. Until such time, the inverters shall be capable of operating at a fixed power factor value between 0.95 lead and 0.95 lag as specified by the Company. The value is supplied below:

1. Operate inverters at a leading power factor ("PF") of **(0.99)**

Note: In the future, the company reserves the right to issue new fixed power factor setting requirements (0.95 lead to 0.95 lag) if necessary.

It is the responsibility of the owner to secure the inverter from any unauthorized access (including physical and remote access) which could alter settings or adversely affect the inverter's ability to operate as required. Security measures should include utilizing secure password settings and/or physical locks on cabinet doors.

ACE will require the capability to remotely disconnect the generator from the grid by communication from its System Operations facility. Such disconnection may be facilitated by either a generator breaker, line recloser, or other method depending upon the specific circumstances and the evaluation of the Company.

A mutually acceptable means of interrupting and disconnecting the generator with a visible break, able to be tagged and locked out, shall be worked out with Company Distribution Engineering.

Special Operating Requirements

1. ACE will require the capability to remotely disconnect the generator from the grid by communication from its System Operations facility. Such disconnection may be facilitated by a generator breaker, or other method depending upon the specific circumstances and the evaluation by ACE.
2. ACE reserves the right to charge the Interconnection Customer operation and maintenance expenses to maintain the Interconnection Customer attachment facilities, including metering and telecommunications facilities, owned by ACE.

Summer Peak Analysis - 2020

Transmission Network Impacts

Potential transmission network impacts are as follows:

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Steady-State Voltage Requirements

No issues identified.

Short Circuit

No issues identified.

Stability Analysis

No issues identified. See Attachment 2 for full report.

Light Load Analysis - 2020

Light Load Studies to be conducted during later study phases (as required by PJM Manual 14B).

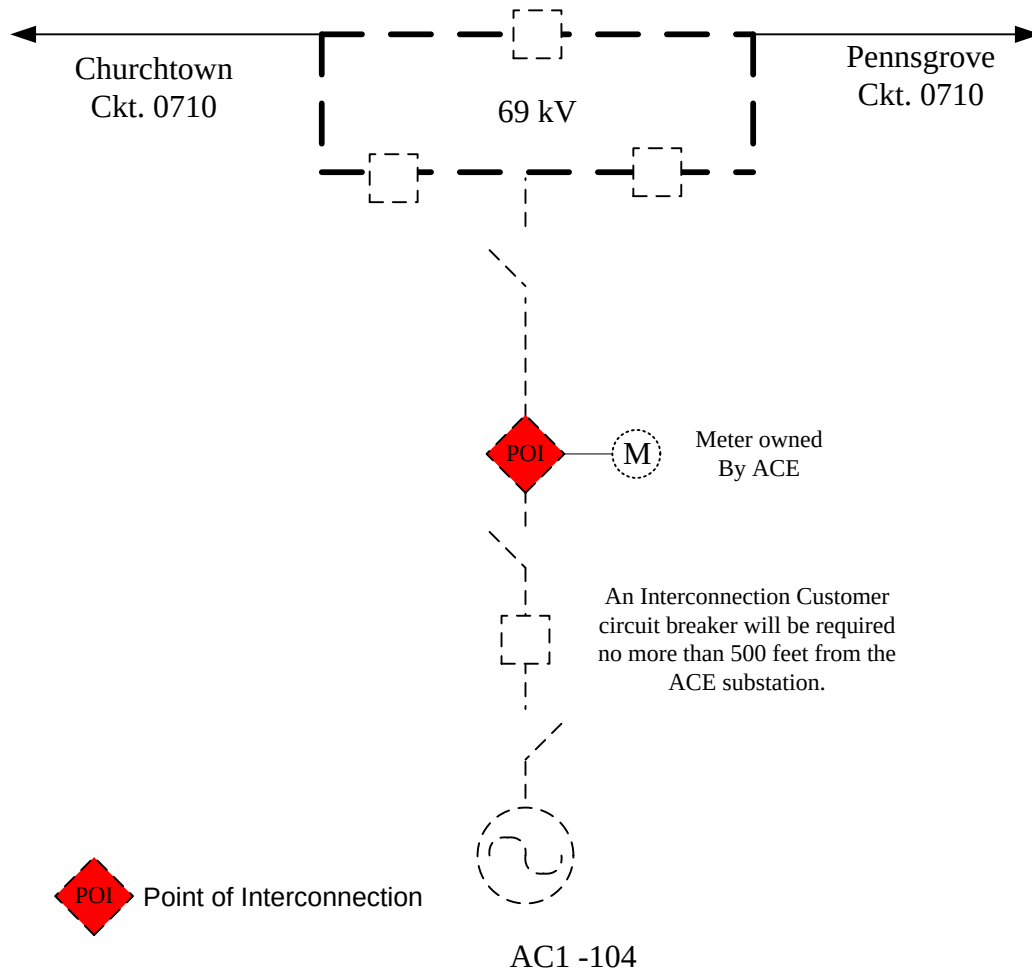
Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Churchtown - Pennsgrove 69 kV AC1-104



Attachment 2

AC1-104
System Impact Study

Dynamic Simulation Analysis

Rev. 0
July 31st, 2017

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Executive Summary

Generator Interconnection Request AC1-104 is for a 19.8 MW Maximum Facility Output (MFO) solar powered generating facility with a Point of Interconnection (POI) at a tap of

the Churchtown - Pennsgrove 69 kV circuit in the Atlantic City Electric Company (AEC) system, Salem County, New Jersey.

The load flow scenario for the analysis was based on the RTEP 2020 Summer Peak case, modified to include applicable queue projects. AC1-104 has been dispatched online at maximum power output, with approximately unity power factor at the POI.

The AC1-104 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

No relevant High Speed Reclosing (HSR) contingencies were identified.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the remaining fault contingencies tested on the 2020 Summer Peak case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 4% for local modes and 3% for inter-area modes.
- b) The AC1-104 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

This project may need to install extra capacitors to fulfill the +/- 0.95 power factor requirement at the POI. Other than that, no mitigations were found to be required.

1. Introduction

Generator Interconnection Request AC1-104 is for a 19.8 MW Maximum Facility Output (MFO) solar powered generating facility with a Point of Interconnection (POI) at a tap of the Churchtown - Pennsgrove 69 kV circuit in the Atlantic City Electric Company (AEC) system, Salem County, New Jersey.

This analysis is effectively a screening study to determine whether the addition of AC1-104 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AC1-104 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

2. Description of Project

Generator Interconnection Request AC1-104 is for a 19.8 MW Maximum Facility Output (MFO) solar powered generating facility with a Point of Interconnection (POI) at a tap of the Churchtown - Pennsgrove 69 kV circuit in the Atlantic City Electric Company (AEC) system, Salem County, New Jersey.

Figure 1 shows the simplified one-line diagram of the AC1-104 loadflow model. Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AC1-104 loadflow model.

The dynamic model for the AC1-104 plant is based on the SMASC_C131 and SMAPPC_B18 PSS/E user defined model supplied by PJM, as indicated by the developer in the System Impact Study Form.

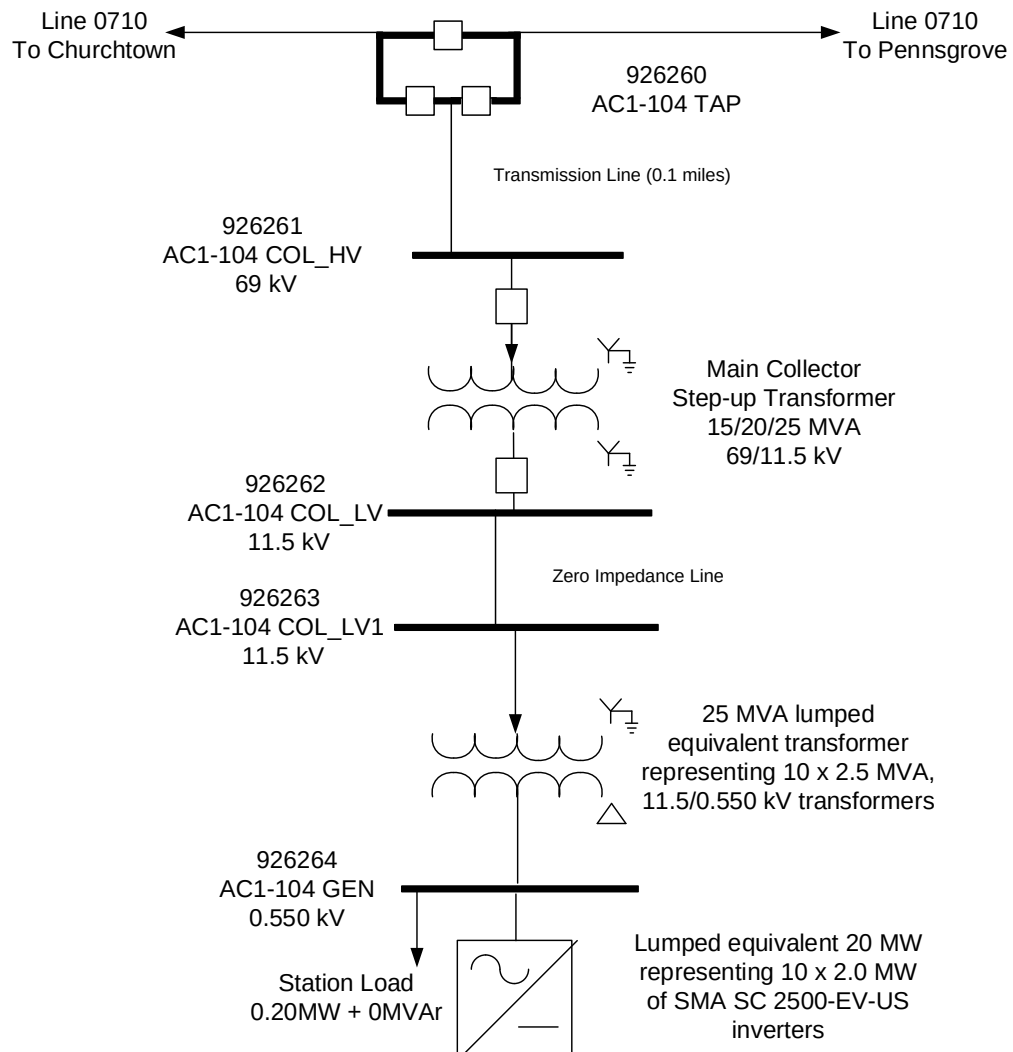


Figure 1: AC1-104 Plant Model

Table 1: AC1-104 Plant Model

	Impact Study Data	Model
Inverters	10 × 2.0 MW SC 2500-EV Solar Inverters MVA base = 2.125 MVA Vt = 0.550 kV	Lumped equivalent representing 10 × 2.00 MW SC 2500-EV Solar Inverters Pgen 20 MW Pmax 20 MW Pmin 0 MW Qgen -6.64 MVAr Qmax 6.64 MVAr Qmin -6.64 MVAr Mbase 21.25 MVA Zsorce 0.0 + j10000.0 pu @Mbase
GSU transformer	10 x 11.5/0.55 kV two winding transformers Transformer base = 2.5 MVA Rating = 2.5 MVA Impedance = 0.007 + j0.057 pu @ MVA base Number of taps = NA Tap step size = N/A	Lumped equivalent representing 10 x 11.5/0.55 kV 2.5 MVA two winding transformers Transformer base = 25.0 MVA Rating = 25.0 MVA Impedance = 0.007 + j0.057 pu @ MVA base Number of taps = 5 Tap step size = 2.5%
Collector step-up transformer	1 x 69/11.5 kV two winding transformers Transformer base = 15 MVA Rating = 15 / 20 / 25 MVA Impedance = 0.004 + j0.079 pu @ MVA base Number of taps = N/A Tap step size = N/A	1 x 69/11.5 kV two winding transformers Transformer base = 15 MVA Rating = 15 / 20 / 25 MVA Impedance = 0.004 + j0.079 pu @ MVA base Number of taps = 5 Tap step size = 2.5%
Auxiliary load	0 MW + 0 MVAr	Not modeled
Station load	0.20 MW + 0 MVAr	0.20 MW + 0 MVAr modelled on the LV side of GSU transformer
Transmission Line	positive sequence impedance: 0.006 + j0.0015 pu charging susceptance: 0.00003j all impedances @ 100 MVA base	positive sequence impedance: 0.006 + j0.0015 pu charging susceptance: 0.00003j all impedances @ 100 MVA base

3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process¹.

The selected load flow scenario is the RTEP 2020 Summer Peak case with the following modifications:

- a) Addition of all applicable queue projects prior to AC1-104.
- b) Addition of AC1-104 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AC1-104.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AC1-104 initial conditions are listed in Table 2, indicating maximum power output, with AC1-104 regulating to unity power factor at the generator bus.

Table 2: AC1-104 machine initial conditions

Bus	Name	Unit	PGEN (MW)	QGEN (MVAR)	ETERM (p.u.)	POI Voltage (p.u.)
925954	AC1-104 GEN10.6900	1	20.0	1.9	1.0000	1.034

Generation within the vicinity of AC1-104 has been dispatched online at maximum output (P_{MAX}). The dispatch of generation in the vicinity of AC1-104 is given in Attachment 3.

¹ Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

4. Fault Cases

Tables 3 listed the contingencies and results that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above. No relevant High Speed Reclosing (HSR) contingencies were studied.

5. Evaluation Criteria

This study is focused on AC1-104, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AC1-104 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 4% for local modes and 3% for inter-area modes.
- b) The AC1-104 is able to ride through faults (except for faults where protective action trips AC1-104).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 4, with results summarized in Table 3.

Due to the frequency protection was disabled due to the PSSE deficiency in calculating frequencies.

For the fault contingencies tested in this study:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 4% for local modes and 3% for inter-area modes.
- b) The AC1-104 generator was able to withstand all contingencies.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

7. Mitigations

This project may need to install extra capacitors to fulfill the +/- 0.95 power factor requirement at the POI. Other than that, no mitigations were found to be required.

Table 3: Fault list**P0: Steady State**

Fault ID	Duration
P0.00	Steady State 20 sec run

P1: Three Phase Faults with normal clearing

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P1.00	3ph @ AC1-104 Main – AC1-104 POI 69kV line, normal clear loss of AC1-104	9	Stable
P1.01	3ph @ AC1-104 POI – Churchtown 69kV line, normal clear	9	Stable
P1.02	3ph @ AC1-104 POI – Carneys Point – Pennsgrove 69kV, normal clear	9	Stable
P1.03	3ph @ Churchtown 69/230kV TF #1, normal clear	9	Stable
P1.04	3ph @ Churchtown 69/230kV TF #2, normal clear	9	Stable
P1.05	3ph @ Churchtown 69/34.5kV TF, normal clear	9	Stable
P1.06	3ph @ Churchtown 69/12.9kV TF, normal clear	9	Stable
P1.07	3ph @ Churchtown – Woodstown 69kV ckt 1, normal clear	9	Stable
P1.08	3ph @ Churchtown – Woodstown 69kV ckt 2, normal clear	9	Stable
P1.09	3ph @ Churchtown – Quinton 69kV line, normal clear	9	Stable
P1.10	3ph @ Churchtown – Salem 69kV line, normal clear	9	Stable
P1.11	3ph @ Churchtown – Beckett 69kV line, normal clear loss of Beckett loads	9	Stable
P1.12	3ph @ Churchtown – AC1-104 POI 69kV line, normal clear	9	Stable
P1.13	3ph @ River – Mickleton 69kV line, normal clear	9	Stable

P4: SLG Stuck Breaker (SB) Faults at Backup Clearing

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.01	SLG @ AC1-104 POI – Churchtown 69kV, SB @ AC1-104 POI, delayed clear loss of AC1-104	9/24	Stable
P4.02	SLG @ AC1-104 POI – Carneys Point – Pennsgrove 69kV, SB @ AC1-104 POI, delayed clear loss of AC1-104	9/24	Stable
P4.03	SLG @ AC1-104 POI – Churchtown 69kV, SB @ AC1-104 POI, delayed clear loss of AC1-104, AC1-104 POI – Carneys Point – Pennsgrove 69kV	9/24	Stable
P4.04	SLG @ Churchtown – Quinton 69kV, SB @ Churchtown, delayed clear loss of Churchtown 69kV bus tie	9/24	Stable
P4.05	SLG @ Churchtown – Quinton 69kV, SB @ Churchtown, delayed clear loss of Churchtown 230/69kV TF #1	9/24	Stable
P4.06	SLG @ Churchtown 69/230kV TF #1, SB @ Churchtown 69kV, delayed clear loss of Churchtown – Salem 69kV line	9/24	Stable
P4.07	SLG @ Churchtown – Salem 69kV line, SB @ Churchtown, delayed clear loss of Churchtown – Beckett 69kV line, Beckett loads	9/24	Stable
P4.08	SLG @ Churchtown – Beckett 69kV line, SB @ Churchtown, delayed clear loss of Churchtown – Woodstown 69kV line ckt 2	9/24	Stable
P4.09	SLG @ Churchtown – Woodstown 69kV line ckt 1, delayed clear loss of Churchtown 69kV bus tie	9/24	Stable
P4.10	SLG @ Churchtown 69/12kV TF, SB @ Churchtown 69kV, delayed clear loss of Churchtown 69kV bus tie	9/24	Stable
P4.11	SLG @ Churchtown 69/230kV TF #2, SB @ Churchtown 69kV, delayed clear loss of Churchtown – AC1-104 69kV line	9/24	Stable
P4.12	SLG @ Pennsgrove – Oldman 69kV line, SB @ Pennsgrove, delayed clear loss of Pennsgrove – Carneys Point – AC1-104 POI 69kV line	9/24	Stable

P5: SLG Fault with Delayed (Zone 2) Clearing

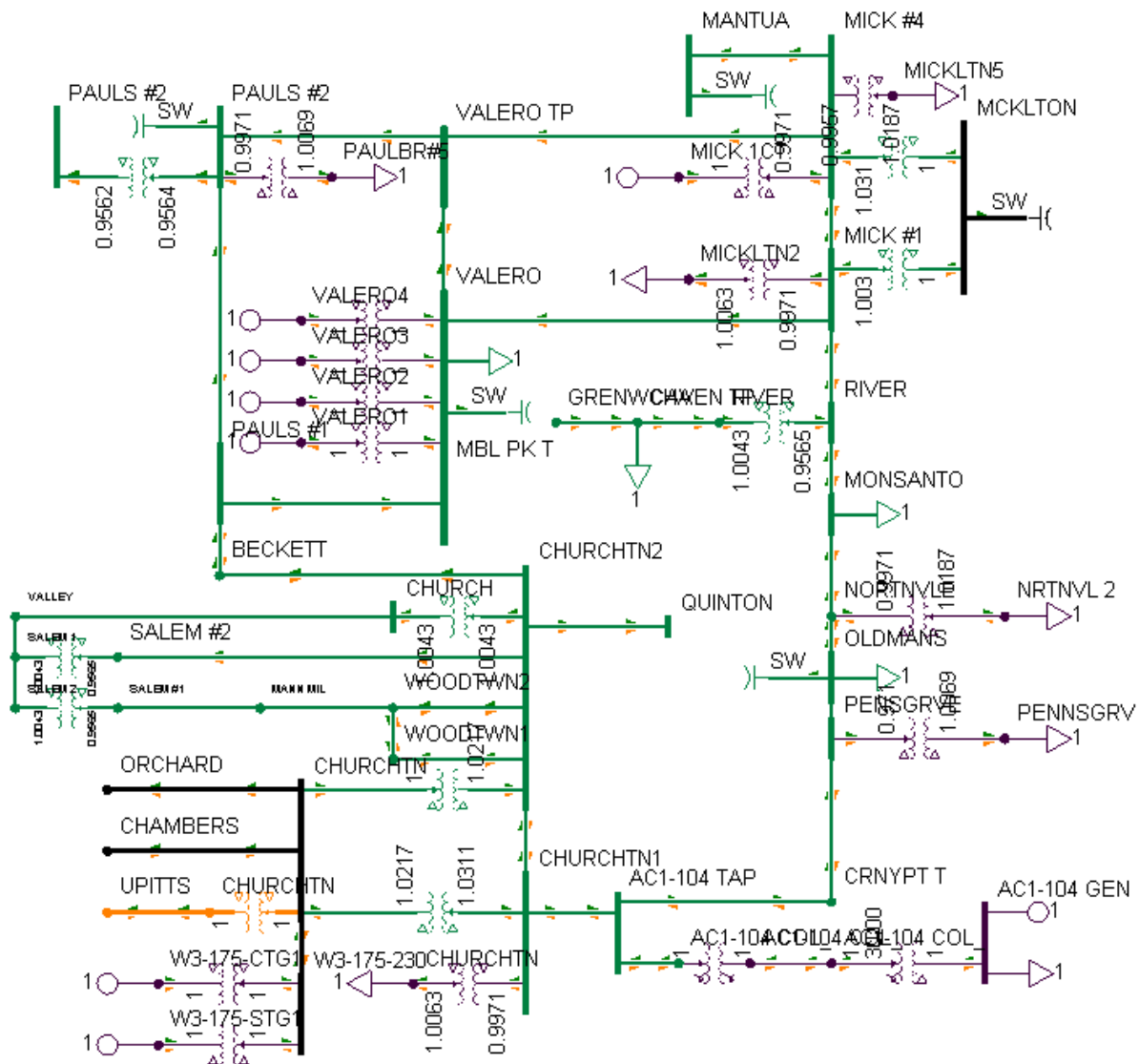
Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
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Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P5.01	SLG @ 80% of AC1-104 POI – Churchtown 69kV, relay failure @ AC1-104 POI, delayed clear	9/42	Stable
P5.02	SLG @ 80% of Churchtown – AC1-104 POI 69kV, relay failure @ Churchtown, delayed clear	9/42	Stable
P5.03	SLG @ 80% of AC1-104 POI – Carneys Point – Pennsgrove 69kV, relay failure @ AC1-104 POI, delayed clear	9/42	Stable
P5.04	SLG @ 80% of Pennsgrove – Carneys Point – AC1-104 POI 69kV, relay failure @ Pennsgrove, delayed clear	9/42	Stable
P5.05	SLG @ 80% of Pennsgrove – Oldman 69kV, relay failure @ Pennsgrove, delayed clear	9/42	Stable
P5.06	SLG @ 80% of Oldman – Pennsgrove 69kV, relay failure @ Oldman, delayed clear	9/42	Stable

P7: Common Structure

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P7.01	CONTINGENCY 'AE11TOWER'	9	Stable
P7.02	CONTINGENCY 'AE1TOWER'	9	Stable
P7.03	CONTINGENCY 'AE9TOWER'	9	Stable
P7.04	CONTINGENCY 'W2275_O2241' /* DOUBLE CIRCUIT TOWER W-2275(MICKLETON - DEPTFORD) AND O-2241(MICKLETON - THOROFARE)	9	Stable

Attachment 1. PSS/E Model One Line Diagram



Attachment 2. AC1-104 PSS/E Dynamic Model

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/*****
****
/**** Project:    AC1-104 - 19.8MW MFO
/**** POI:       69 kV bus (926260) at a tap of the Churchtown - Pennsgrove 69 kV
/**** Inverter:   SC 2500-EV
/**** Size:      10 x 2.0 MW Solar PV
/**** PSSE Version: 33
/****
****
/
/
926264 'USRMDL' 1 'SMAPPC18' 4 0 4 28 9 21 926263 926262 1 0
          3 0.5 0.25 0.04 0.0 0.1 1.0 0.04 0
          0 1 0 10 -0.001
          0.1 1.0 0.04 0 1 0.2 0.2
          0.8 0.91 0.915 1.060 1.055 0.3 1.0/
/          CtlMod KP_PF KI_PF PFXdcrTm PFDB KP_Vol KI_Vol VolXdcrTm VolDB
/          PFNomTot VolNomTot QNomTot VolDroop HybCtlTun
/          KP_P KI_P PXdcrTm PDB PNomTot QCommTm PCommTm
/          PFLim FRTThrhLo1 FRTThrhLo2 FRTThrhHi1 FRTThrhHi2 FRTHldTm Reserved
926264 'USRMDL' 1 'SMASC131' 1 1 0 76 15 193
          1.0 1.0 0.0 0.8 1.0 0.0 1.0 1.0 0.35
          1.0 0.0 5.0 0.5 2.0
          0.0 0.9 0.5 1.0 0.9 0.9
          0.0 0.2 0.05 0.4
          0.35 0.35 1.0 2.0 2.0 2.0 0.1 0.098 0.1 0.098 0.2 0.2
          0.01 1.0 20.0 0.5 0.61 0.62 0.0 0.125 0.1
          1.2 0.1 1.18 1.0 1.15 2.0 0.88 12.0 0.6 5.0 0.5 3.0
          65.0 0.1 64.0 1.0 61.5 3.0 59.3 5.0 57.0 3.0 50.0 0.1
          30.0 0.0 10.0 30.0 0.0 1.0 1.0/
/          PPrim PWNom QVArNom PFLIM PFPF PFPFExt QVArMod QoDEna QoDQMax
/          VArCtlVol_Volref VArCtlVol_VolDB VArCtlVol_VArGra VArCtlVol_VArMax
VArCtlVol_VArTm
/          PFPFExtStr PFPFStr PFWStr PFPFExtStop PFPFStop PFWStop
/          WCtlHzMod PHzStr PHzStop PWGra
/          WGra VArGra DGSMOD DGSArGraNom DGSArGraNomHi DGSArGraNomLo
DbVolNomMax DbVolNomMaxH DbVolNomMin DbVolNomMinH DGSqRcvrTm
DGSNqRcvrTm
/          VArCmdFltTm FRTPreErrVena FRTPreErrTm FRTSwOffTm FRTThrhld1
FRTThrhld2 VCtlCharEna VCtlCharTm VCtlCorTm
/          VCtlMax VCtlMaxTm VCtlhhLim VCtlhhLimTm VCtlhLim VCtlhLimTm
VCtlLim VCtlLimTm VCtlLim VCtlLimTm VCtlMin VCtlMinTm

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/ HzCtlMax HzCtlMaxTm HzCtlhhLim HzCtlhhLimTm HzCtlhLim HzCtlhLimTm
HzCtlllLim HzCtlllLimTm HzCtlllLim HzCtlllLimTm HzCtlMin HzCtlMinTm
/ KPLL1 PLLFlag KPPLL2 KIPLL2 Reserved Reserved GenTrpFlag

Attachment 3. AC1-104 PSS/E Case Dispatch

Bus Number	Bus Name	Id	In Service	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)
200036	SALEM-G1 22.000	1	1	1253	1253	1018	229.4	535	100
200037	SALEM-G2 22.000	1	1	1245	1245	1008	229.4	542.9	100
200039	HOPE CG1 22.000	1	1	1320	1320	0	560	560	108
200062	SALEM G3 22.000	1	1	38.4	38.4	0	14	14	-14
227999	CARDSVC 16.000	1	1	0	0	0	80.54	150	-100
228008	CEDSVC 16.000	1	1	0	0	0	6.309	150	-100
228200	CARL#1CT 13.800	1	1	38.7	38.7	0	-1.3	3.575	-1.3
228201	CARL#2CT 13.800	1	1	40.5	40.5	0	-4.27	8	-5
228202	CUMB CT 13.800	1	1	93	93	90.8	3.252	18	0
228203	P06 13.800	1	1	92	92	92	-14.27	100	-55
228206	SHRMN CT 13.800	1	1	93	93	81	-4.292	39	-6
228231	DENNSVC 16.000	1	1	0	0	0	49.78	150	-100
228251	CARLLS#4 12.900	1	1	4	4	0	0	0	0
228304	LOGAN 22.000	1	1	219	219	50	52.45	63.67	-58.2
228306	PCLP STM 13.800	1	1	60.2	60.2	0	16.77	45	-59
228307	PCLP GT 13.800	1	1	60.1	60.1	0	15.26	29.25	-25.05
228309	CCLP NUG 23.000	1	1	240	240	46	92.67	92.67	-94.6
228334	MANNMILG 4.1600	1	1	2.8	2.8	0	-1.016	2.8	-2.8
228334	MANNMILG 4.1600	2	1	2.8	2.8	0	-1.016	2.8	-2.8
228334	MANNMILG 4.1600	3	1	2.8	2.8	0	-1.016	2.8	-2.8
228343	QUINTN#1 12.900	1	1	2	2	0	0	0	0
228400	MICK 1CT 13.800	1	1	73	73	53	0	29	0
228471	VALERO1 13.800	1	1	22.1	22.1	0	-7.312	15.65	-11.74
228472	VALERO2 13.800	1	1	10.7	10.7	0	-5.681	7.575	-5.681
228473	VALERO3 13.800	1	1	10.7	10.7	0	-3.49	7.575	-5.681
228484	VALERO4 13.800	1	1	13	13	0	-3.266	9.204	-6.903

297082	V2-035 C 12.900	1	1	0.76	0.76	0	0	0	0
297083	V2-035 E 12.900	1	1	1.24	1.24	0	0	0	0
297103	V2-046 C 12.900	1	1	7	7	0	0	0	0
297104	V2-046 E 12.900	1	1	12.9	12.9	0	0	0	0
902091	W1-130C 12.900	1	1	3.8	3.8	0	0	0	0
903963	W3-175-STG1 18.000	1	1	217	217	0	5.931	175	-120
903965	W3-175-CTG1 25.000	1	1	330	330	0	5.931	255	-160
904531	V4-054 C 12.900	1	1	7.6	7.6	0	0	0	0
905271	W4-027 C 12.900	1	1	3	3	0	0	0	0
909091	X2-027 C OP169.000	1	1	0.76	0.76	0	0	0	0
909092	X2-027 E OP169.000	1	1	1.24	1.24	0	0	0	0
924531	AB2-102 GEN 18.000	1	1	236.5	236.5	0	96.32	140	-100
926264	AC1-104 GEN 0.5500	1	1	20	20	0	-4.526	6.64	-6.64
931191	AB1-169A- GEN20.500	1	1	582	582	0	53.55	335	-230